



Modeling influenza dynamics in Italy: a multi-stage SIRS approach with integrated vaccination

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Outline

- Epidemiological context and project objectives
- Classic SIR/SIRS models and their limitations
- Proposed SIRnS model
- Data used and numerical implementation
- Main results on the 2017-2018 case study
- Parameter sensitivity analysis
- Conclusions

Context and Objective

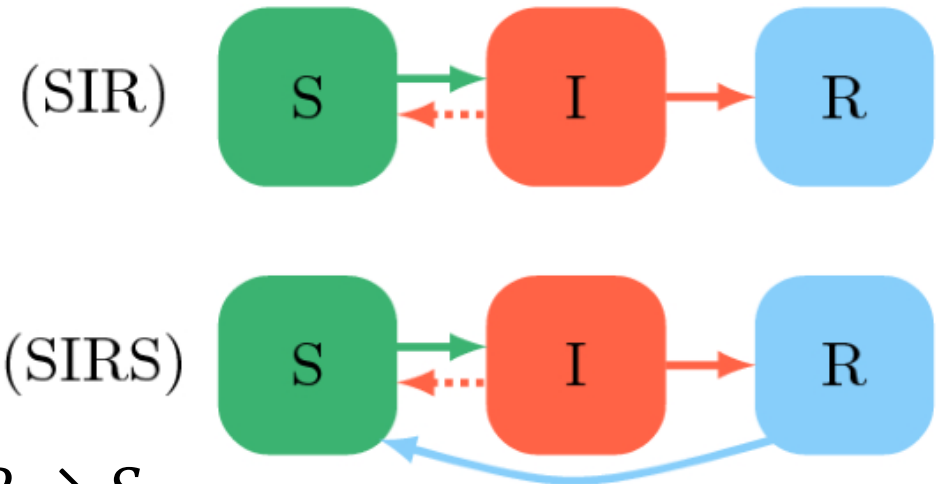
- This project focuses on seasonal influenza.
- Seasonal influenza is characterized by:
 - Rapid transmission
 - Temporary immunity
 - Annual vaccination campaigns
- Objective: to analyze infections in Italy using an n-level structure for immunity, integrating vaccination as a dynamic flow rather than a static compartment.

SIR and SIRS models

Compartments:

- S: susceptible
- I: infected
- R: recovered/immune

In the SIRS model, immunity is not permanent $R \rightarrow S$



The following constraints apply to both models:

- Constraint 1: Each individual in the population must be assigned to only one of these classes.
- Constraint 2: We assume that every infected individual is infectious.
- Constraint 3: The number of individuals N in a population is fixed.

$$N = S(t) + I(t) + R(t)$$

SIRnS model

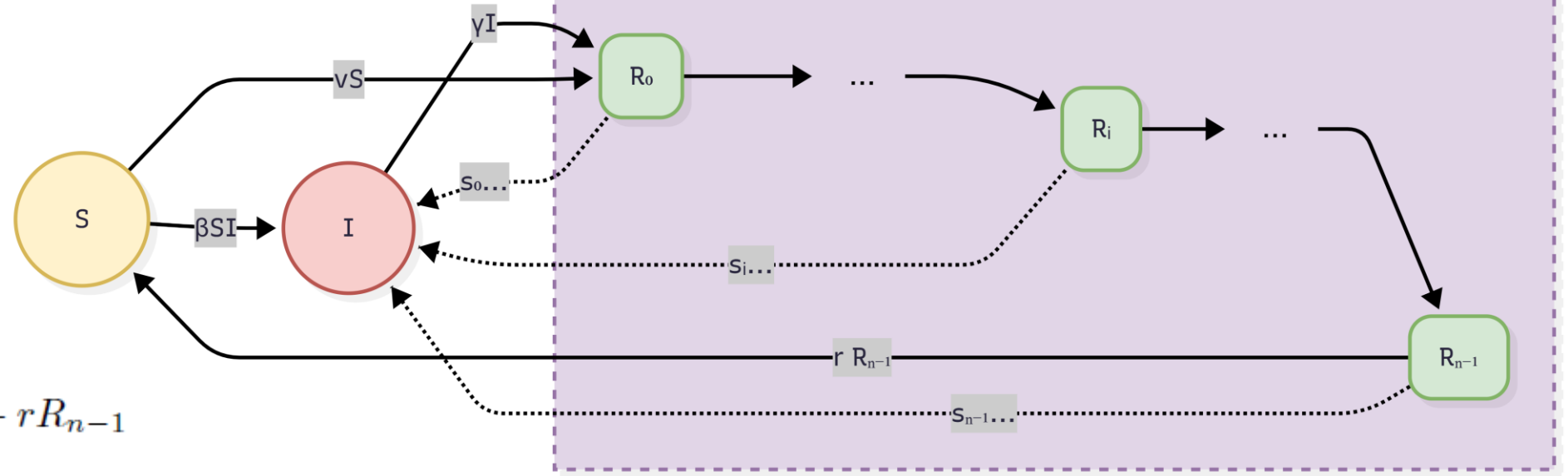
- Extension of the classic SIRS model
- The R compartment is divided into n subcompartments:
 - $R_0, R_1, \dots, R_{n-1}$
- Each subcompartment represents a different level of susceptibility
 - With $s_0 < s_1 < \dots < s_{n-1}$
 - Where $s_i \in [0, 1]$
- This allows us to model:
 - Gradual decline in immunity



SIRnS model

- No dedicated vaccine compartment
- Reason:
 - The flu vaccine does not guarantee complete immunity.
 - The aim is to also model the decline in immunity induced by vaccination.
- Vaccinations are modeled like the flow:
$$S \rightarrow R_0$$
- The intensity of the flow is proportional to:
 - The actual coverage rate
 - The duration of the vaccination campaign

SIRnS model



$$\begin{cases} \frac{dS}{dt} = -\beta SI - vS + rR_{n-1} \\ \frac{dI}{dt} = \beta SI + \sum_{i=0}^{n-1} (s_i \beta R_i I) - \gamma I \\ \frac{dR_0}{dt} = \gamma I + vS - s_0 \beta R_0 I - rR_0 \\ \frac{dR_i}{dt} = rR_{i-1} - s_i \beta R_i I - rR_i \quad (0 < i < n) \end{cases}$$

β = infection rate.

γ = recovery rate.

v = vaccination rate.

s_i = residual susceptibility subcompartment R_i

r = immunity decay rate.

Numerical implementation

- The ODE system is solved numerically
- Programming language: Python
- Main libraries:
 - NumPy and Pandas for data handling
 - SciPy for ODE integration
 - Matplotlib for visualization
- Simulations are performed over:
 - A single influenza season (November to April)



Data used and calibration

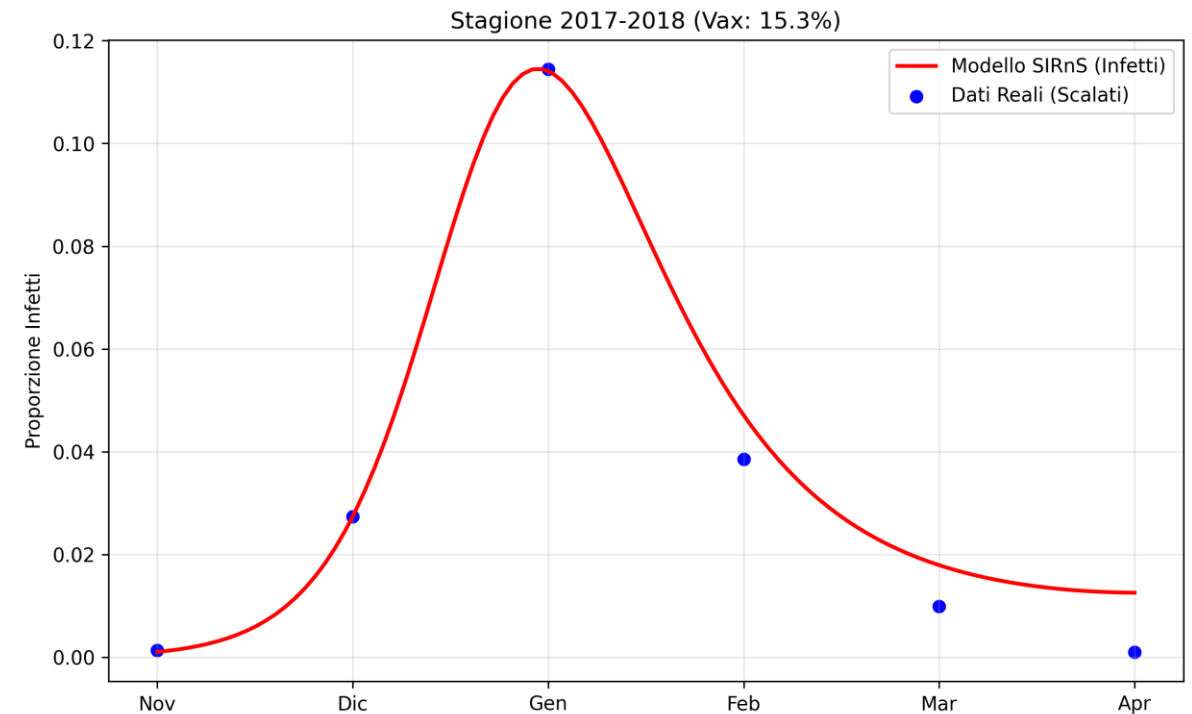
- Epidemiological data:
 - FluNet (WHO)
 - Weekly cases aggregated monthly
 - Only available for November-April
- Vaccination data:
 - Ministry of Health
- Case study: 2017-2018 season in Italy

Basics Parameters:

β (Infection rate)	8
γ (recovery rate)	4
Rip0 β/γ (Basic reproduction)	2
r (immunity decay rate)	0.5
n (Stages of immunity)	3
t_{vax} (duration vaccination campaign)	2

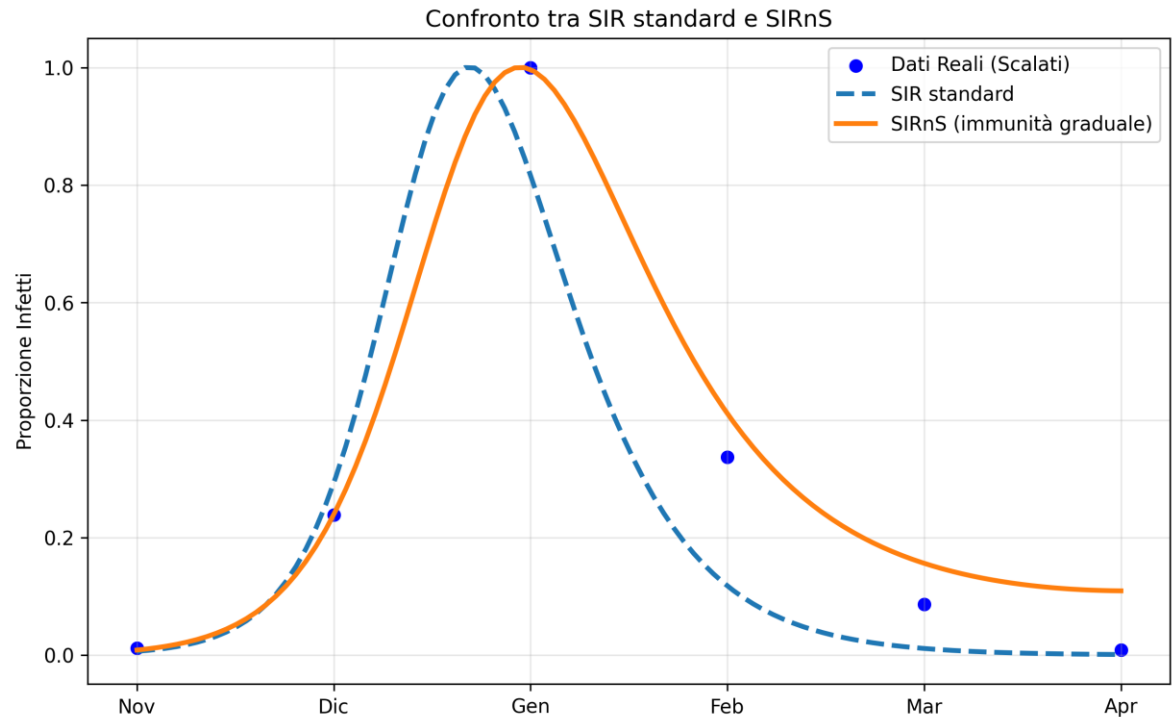
Comparison between model and real data

- The comparison focuses on:
 - Temporal position of the peak
 - Overall shape of the curve
- The actual data is scaled for visual comparison
- The model correctly reproduces the timing of the peak, and the growth phase is consistent with the model



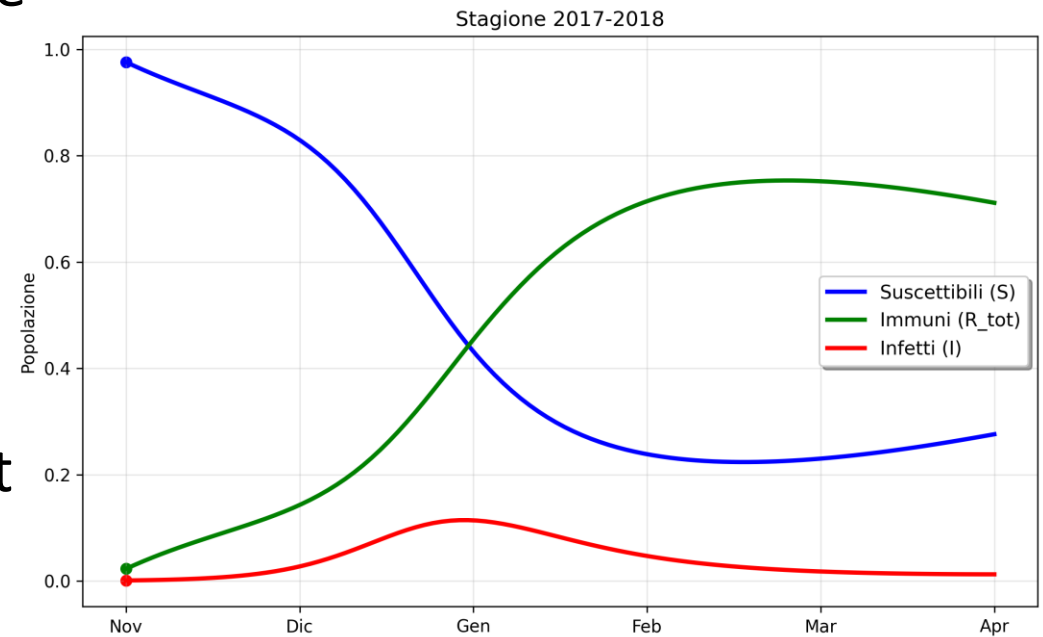
Comparison between SIR and SIRnS

- Both have been normalized to compare the shape of the trend
- The SIRnS shows:
 - Slower growth and decline
 - More realistic epidemic decline
- The standard SIR shows:
 - A premature peak
 - Very steep decline



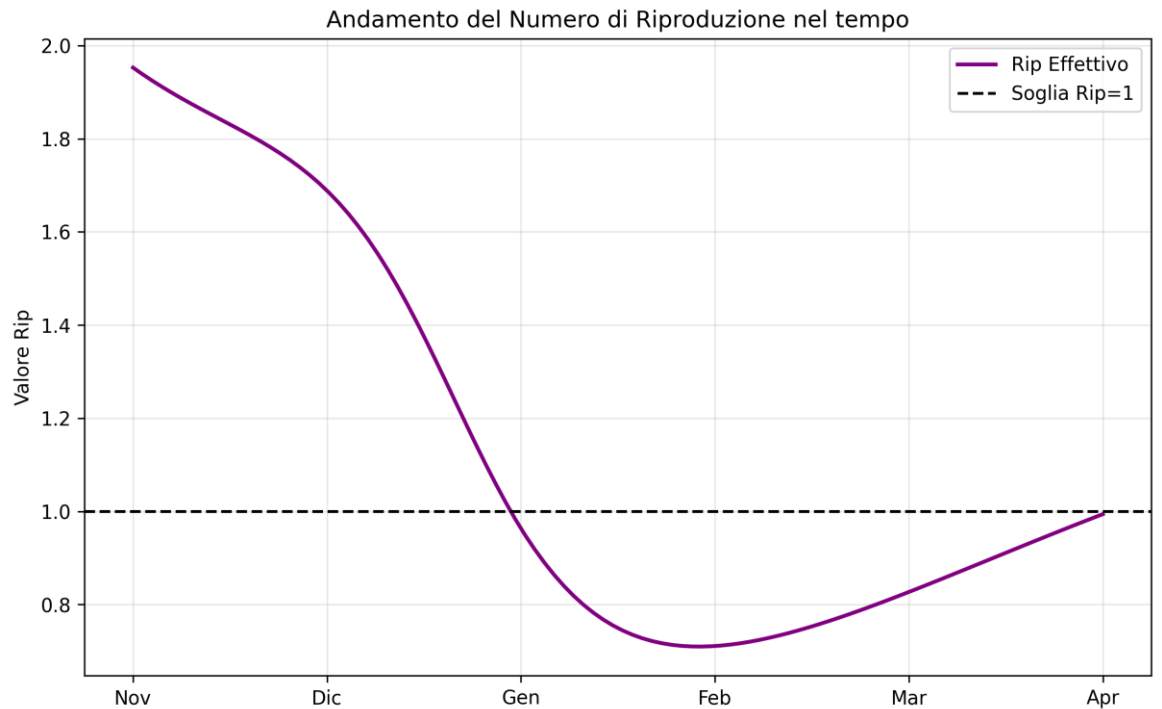
Population dynamics

- The curve of susceptible and that of immune intersect in January.
- The peak of infected corresponds to the moment when the number of immune exceeds that of susceptible individuals.
- It is estimated that approximately 15% of the vaccination campaign will be carried out in the months leading up to November.
 - Susceptible individuals do not start from 1.
 - Immune individuals do not start from 0.



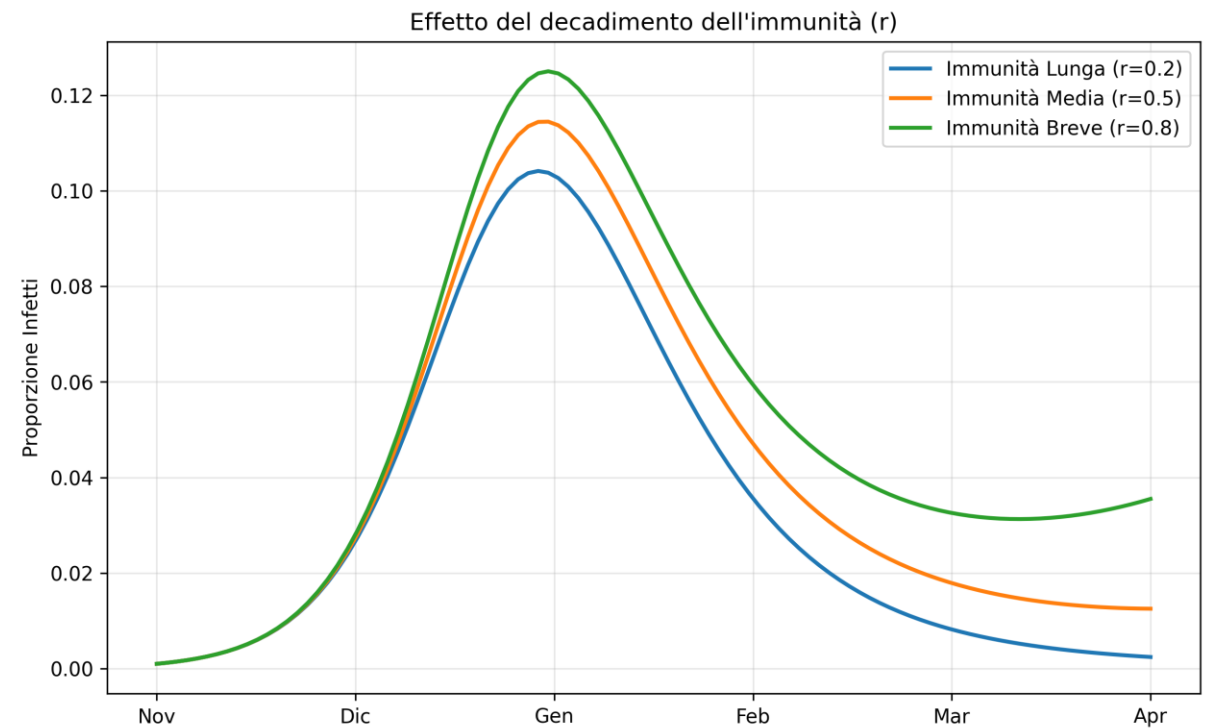
Analysis of the reproduction number

- It indicates how many new infections an infected individual generates on average.
- R_0 calculated as β/γ is equal to 2.
- R varies over time because it is multiplied by the number of susceptible individuals at time t .
- Epidemiological interpretation:
 - $R(t) > 1$: the epidemic grows
 - $R(t) = 1$: equilibrium level
 - $R(t) < 1$: the epidemic dies out



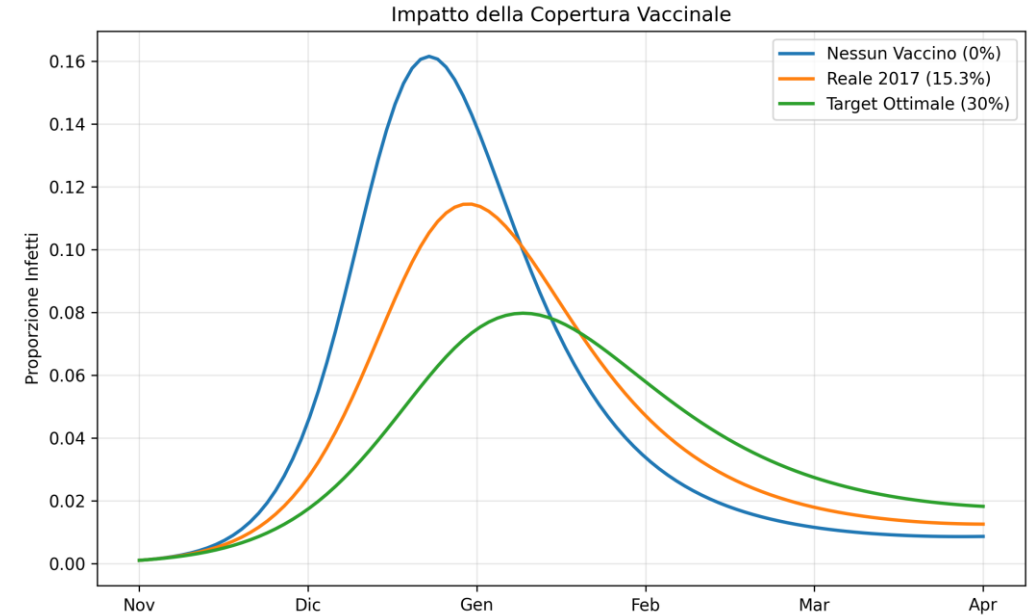
Waning immunity

- The effect of the parameter r , which controls the rate of loss of immune protection, is analyzed.
- High values of r lead to:
 - Higher endemic peaks
 - More persistent epidemic tail
- Low values of r lead to:
 - Reduced peak intensity
 - Accelerated extinction of the epidemic



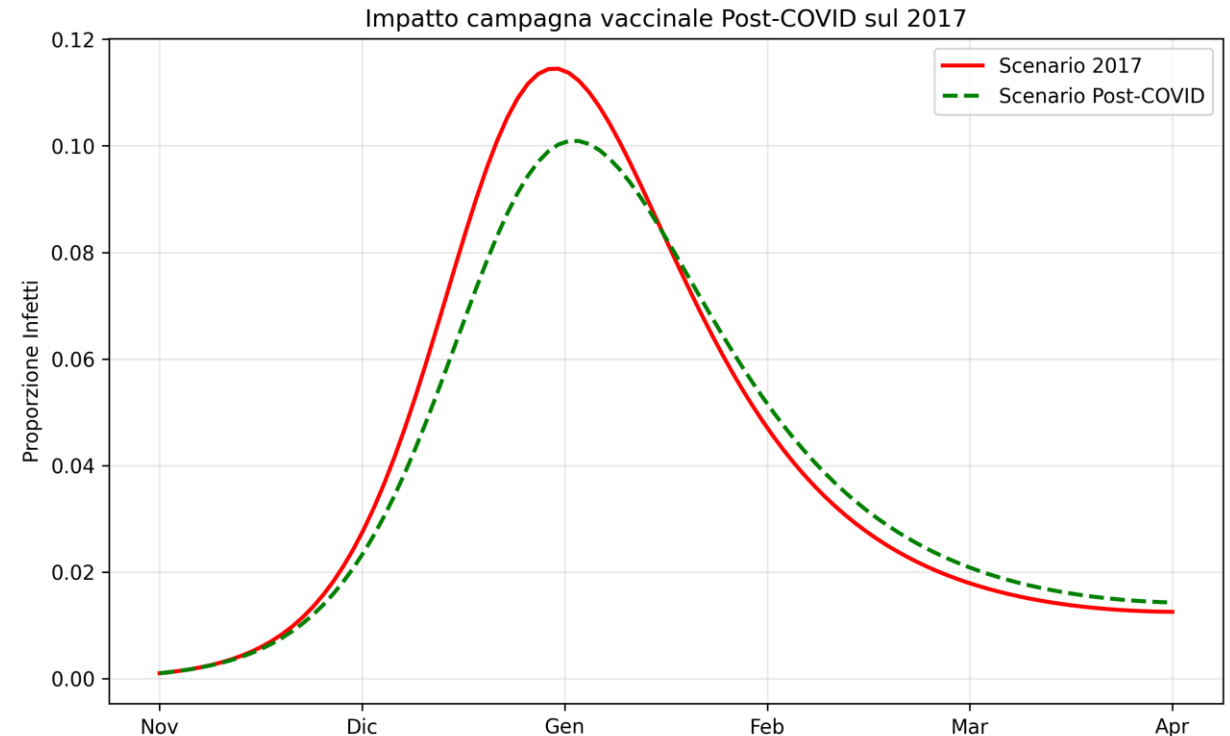
Impact of vaccination coverage

- Comparison of 3 scenarios
 - No vaccination (0%)
 - Actual coverage in 2017 (15.3%)
 - Optimal scenario (30%)
- The absence of vaccination leads to an earlier and higher peak.
- Increased vaccination coverage slows the spread of infection and reduces the peak.



Post-COVID vaccination rates

- What would have happened in 2017 if we had had the same vaccination rate as in 2021 (20.5%)?
 - The peak would have been lower, at 11.8%, and slightly delayed.
- In the post-COVID years, the vaccination campaign has been more intense, leading to higher vaccination rates and therefore lower epidemiological peaks.





Conclusions

- Validation of the SIRnS model:
 - The multi-stage model ($n=3$) approximates the actual data for the 2017-18 season better than the classic SIR model.
 - It correctly captures the asymmetry of the epidemic curve and the slow decline due to waning immunity.
- Immunity dynamics:
 - Partial susceptibility and immunity decay are the main components of virus persistence at the end of the season.
- Impact of vaccination:
 - Applying 2021 vaccination rates, the model estimates a 9% reduction in the peak, confirming the importance of broad coverage.



THANK YOU FOR THE ATTENTION!